

Literature Survey of Patient-Controlled EHR and Consent Platforms (2018–2025)

We reviewed recent academic and industry work on patient-centric health records and consent-driven data sharing. For each key reference we list Title, Authors, Year, Publisher, DOI and summarize the problem addressed, methods, findings, limitations, and relevance to ANUJNA. Citations are provided for quoted material or key facts.

- **“ACTION-EHR: Patient-Centric Blockchain-Based Electronic Health Record Data Management for Cancer Care”** – Dubovitskaya *et al.*, 2020, *Journal of Medical Internet Research* (DOI:10.2196/13598) [6†L250-L259] [6†L262-L266] .

Problem: EHR data sharing is slow, manual, and lacks patient control; centralized HIEs face security and privacy issues. Patients have legal rights (e.g. under HIPAA/GDPR) to manage consent, but existing systems don't support this well [6†L291-L300] [8†L818-L827] .

Methodology: Developed a prototype *ACTION-EHR* system on Hyperledger Fabric (permissioned blockchain). Used HL7 FHIR to represent patient data, with off-chain encrypted storage and on-chain metadata/permissions. Patients control access via smart contracts; data-sharing transactions were exposed through REST APIs. Deidentified cancer patient data at Stony Brook were used for testing [6†L250-L259] .

Key Findings: The prototype demonstrated feasibility of a blockchain-based, patient-centric sharing platform. It enabled patients to manage cross-hospital records securely, with fine-grained access control and audit trails. It showed millisecond-scale performance for permission changes and maintained data integrity and confidentiality [6†L250-L259] [8†L818-L827] .

Limitations: The authors note challenges: the current prototype lacks a “break-glass” emergency-access mechanism, and immutability conflicts with GDPR’s “right to be forgotten” [8†L818-L827] . Single-point failures (e.g. if only one ordering node is used) must be addressed in deployment. Some solutions (group signatures, redactable blockchain) are suggested for future work [8†L818-L827] .

Relevance to ANUJNA: ACTION-EHR exemplifies a patient-controlled sharing platform similar in vision to ANUJNA (patient-managed consent, HL7 FHIR integration, auditability) [6†L250-L259] . It underscores the benefits of permissioned blockchain and FHIR for interoperability. ANUJNA could build on this by adding explicit “break-glass” and multi-regulatory compliance features.

- **“Strengthen Electronic Health Records System (EHR-S) Access-Control to Cope with GDPR Explicit Consent”** – Carvalho Junior & Bandiera-Paiva, 2020, *Journal of Medical Systems* (DOI:10.1007/s10916-020-01631-5) [21†L132-L141] [21†L142-L150] .

Problem: Most EHR systems lack mechanisms for explicit patient consent. Under GDPR and similar laws, individuals must control their health data, but conventional Role-Based Access Control (RBAC) models in EHRs only grant control to administrators or roles. Patients rarely can decide how their own records are shared [21†L132-L141] .

Methodology: The authors propose an adaptation of RBAC that incorporates *patient-centric control*. They extend standard RBAC by adding patient-managed permissions. In their model, the patient becomes the sole “information owner” and can encrypt their data; group memberships and an

existing permission matrix (PERMS) are leveraged to vet access [21†L132-L141] [21†L142-L150] . The approach mimics Discretionary Access Control (DAC) within the RBAC framework.

Key Findings: The paper shows that extending RBAC with patient-run encryption and consent capabilities can give patients exclusive control over their records [21†L132-L141] . Patients can grant or revoke access to providers via consent artifacts (following the MeitY/NDHM framework). The design enables granular, patient-controlled permissions.

Limitations: This work is largely theoretical/design-focused. There is no deployed prototype or performance evaluation. It does not address emergency access (“break-glass”) or interoperability standards. It assumes healthcare systems will integrate the patient-as-owner logic into their RBAC implementations.

Relevance to ANUJNA: Carvalho’s model reinforces ANUJNA’s patient-centric access control: treating patients as data owners who grant consent. ANUJNA’s use of workflows and audit logs goes beyond this proposal. A difference is that ANUJNA plans decentralized ledgers for audit, while Carvalho’s design is a traditional RBAC extension. This suggests ANUJNA’s blockchain approach is a potential research/engineering improvement over pure RBAC enhancements [21†L132-L141] .

- **“Generic Patient-Centered Blockchain-Based EHR Management System”** – Haddad *et al.*, 2023, *Applied Sciences* (DOI:10.3390/app13031761) [52†L533-L542] [52†L543-L552] .

Problem: As in ACTION-EHR, patient data are scattered and sharing is often insecure/inefficient. Centralized EHRs pose privacy risks. There is a need for systems enabling patients to *grant or revoke* access to their records across multiple stakeholders [52†L527-L536] .

Methodology: Developed *PCEHRM*, a blockchain (Ethereum) + IPFS-based system. Patient medical records (or hashes) are stored on IPFS; permissions and metadata are on Ethereum. An Ethereum smart contract – a “patient-centric access control protocol” – enforces distributed access control. A web UI allows patients and providers to initiate sharing transactions. The system was prototyped with Truffle/Web3 on Ethereum.

Key Findings: The authors demonstrate that a decentralized architecture can support patient-managed EHR sharing. Patients (and stakeholders like labs, researchers) control data sharing via the smart contract. Performance tests (varying peers and file sizes) show the approach is **practicable and efficient** [52†L533-L542] [52†L543-L552] . For example, blockchain immutability and distribution were leveraged for security/privacy; the prototype’s timing and cost metrics were evaluated and found acceptable for moderate data sizes [52†L549-L554] .

Limitations: Being Ethereum-based, the approach may face high on-chain costs and latency for large-scale use. The system assumes a public blockchain (Ethereum) – in practice a permissioned chain would be required for healthcare. The design also lacks explicit “break-glass” or compliance mechanisms. It was tested only in simulation; scalability beyond the prototype was limited (they note IPFS costs, Ethereum throughput).

Relevance to ANUJNA: This work is highly similar in goal (patient-controlled record management). It differs technically (Ethereum vs. ANUJNA’s envisioned permissioned blockchain) and architecture (it uses IPFS for data, while ANUJNA may use cloud + metadata on chain). Its emphasis on *patient grant/revoke* and on-chain logging is in line with ANUJNA’s vision. The smart-contract access protocol could inspire ANUJNA’s workflow logic. However, using Ethereum/PCEHRM’s public chain is an engineering choice; ANUJNA uses permissioned for performance. Thus this is more an incremental validation of patient-blockchain ideas than a novel research insight.

- **“Blockchain-Enabled Next-Generation Access Control for Personal Health Records”** (*Lecture Notes in Networks and Systems vol. 425*) – Dong *et al.*, 2022 (cited in Dong *et al.* 2023 Heliyon)

【27†L325-L332】 【27†L334-L342】 . *(This is their prior conference work underlying the 2023 Heliyon paper below.)*

Problem: U.S. lacks a longitudinal personal health record (PHR) system outside the VA. EHRs remain isolated by institutions and rarely shared due to trust, privacy, business concerns. Patients distrust that only institutions (not themselves) manage record sharing 【27†L298-L307】 .

Methodology: Proposed a *Blockchain-enabled Access Control (BAC)* model. Patients onboard to a platform (“Patient Health Information Exchange”) where multiple providers join. The BAC model uses Hyperledger Fabric. Trust is rooted in patient consent: healthcare orgs join only after patient validation. They built a web-based prototype with one patient and two provider nodes (Fabric network). Patients control their PHR via a “Blockchain-Enabled Smart Token” (BEST) system for permission records.

Key Findings: The system lets patients manage consent granularity and links EHRs from different providers. Performance tests show sub-second permission-change latency even at scale. They report: “We constructed a Web-based PHR sharing platform with patient-controlled security, privacy, and consent granularity... The prototypical instantiation suggested the feasibility of combining blockchain with next-gen access control to tackle a longstanding PHR problem” 【27†L325-L332】

【27†L334-L342】 . Key attributes achieved include auditability, scalability, distributedness, *patient consent autonomy*, and “zero-trust” architecture 【27†L334-L342】 . **Limitations:** Their evaluation was limited to synthetic data and a small network. They note that using a non-graph database hindered scalability and suggest a graph DB for future work 【27†L325-L332】 . No user studies or real patient data were used. Emergency access (“break-glass”) is not discussed. **Relevance to ANUJNA:** This work is directly in ANUJNA’s domain: patient-mediated PHR sharing with blockchain and web apps. It confirms feasibility of Hyperledger Fabric for such use-cases and highlights performance considerations. ANUJNA’s design overlaps (patient-controlled permissions, audit) but ANUJNA adds a unified health identity and multi-regional scope (India). This is a strong example of existing patient-centric sharing architecture; any novelty in ANUJNA would need to go beyond this model (e.g. more automation, cross-country compliance).

- **“Towards Patient-Centric Healthcare: Leveraging Blockchain for Electronic Health Records”** – Nhan *et al.*, 2024 (SIGMIS-CPR ’24, DOI:10.1145/3632634.3655883) 【13†L550-L558】 【13†L559-L567】 .

Problem: Conventional EHRs are centralized and fragmented. There is growing interest in using blockchain to eliminate single points of failure, enhance security/integrity, and give patients data ownership 【13†L552-L561】 . Despite many conceptual studies, few works fully implement these ideas.

Methodology: Propose a novel EHR framework integrating blockchain and FHIR. Implement a **modular, patient-centric EHR system** with smart contracts and IPFS for storage. They built and tested a web-based prototype on a local blockchain. Key features include decentralized identity (patients login via username/password + blockchain keys), OAuth2/OpenID Connect, and load-testing on an Ethereum-based network.

Key Findings: The paper emphasizes four contributions: a blockchain-FHIR integration framework for data integrity and privacy; a patient-centric model giving individuals full control and access over their data; robust security protocols leveraging blockchain to prevent breaches; and a **modular design** allowing component substitution 【13†L550-L558】 【13†L559-L567】 . They claim their implementation achieves these goals (details not fully in abstract but inferred). They compare to prior work and stress that their system has load testing and real prototype (unlike many theoretical works) 【13†L550-L558】 【13†L559-L567】 . **Limitations:** The paper appears more descriptive of the

system and its architecture; concrete results are not detailed in the excerpt. As a recent conference paper, the novelty is primarily in design and implementation, not in proving new algorithms. Its evaluation is on a test network only. Emergency access is not mentioned.

Relevance to ANUJNA: The goals align: patient empowerment, decentralized security, FHIR interoperability. Nhan *et al.* illustrate a patient login and smart-contract-based sharing system. ANUJNA's differences might include a stronger emphasis on identity federation (ANUJNA is an "Identity Platform") and region-specific policies (GDPR/HIPAA compliance). Both use modern tech (blockchain, IPFS/FHIR). If ANUJNA's design is similar, the research novelty may lie in specific workflows or governance rather than the core idea.

• **"WiraChain: a blockchain and FHIR-based platform for improving clinical data interoperability in healthcare"** – Grau *et al.*, 2026, *Frontiers in Public Health* (DOI:10.3389/fpubh.2026.1746772)

【30†L347-L355】 【30†L356-L362】 【31†L838-L847】 .

Problem: In Peru (and similarly elsewhere), disparate systems and lack of interoperability lead to fragmented records and duplicated tests. Existing blockchain-FHIR proposals are often conceptual; there are few fully implemented systems that include patient consent and real web applications

【30†L342-L351】 .

Methodology: Developed *WiraChain*, a decentralized web app using Hyperledger Besu (blockchain) + HL7 FHIR + a React/.NET stack. Key features: **patient-controlled permission granting/revocation**, standard RBAC authentication, an immutable audit trail (via blockchain), and a REST FHIR API for data exchange. Used hybrid storage: blockchain for authorization/audit, off-chain FHIR repositories for clinical data. Validated with synthetic data and stress tests.

Key Findings: WiraChain "enables patient-controlled permission granting and revocation" and reliably manages permissions and clinical data exchange under load 【30†L347-L355】 . The platform maintained efficient, secure operation in simulated workflows. The authors conclude it effectively bridges the gap from theoretical frameworks to practical applications, illustrating feasibility of combining blockchain with FHIR for **secure, patient-centered, interoperable** health operations

【30†L347-L355】 【30†L356-L362】 . Importantly, WiraChain includes an explicit **emergency ("break-glass") access** mechanism: predefined clinical roles (e.g. ER doctors) can override consent in life-threatening situations; such events are immutably logged on-chain and subject to post-incident audit 【31†L838-L847】 . All on-chain transactions record only minimal metadata (no PHI) for privacy.

Limitations: WiraChain is validated via simulation and architectural evaluation; no live clinical deployment yet. Formal security audits were not done as it's a prototype. The emergency access system is descriptive; user acceptance of such override policies is untested. Being focused on Peru, it assumes compliance with local laws (though designed to align with HIPAA/GDPR principles)

【31†L838-L847】 .

Relevance to ANUJNA: WiraChain is a mature example of a blockchain+FHIR patient-centric system with many features ANUJNA envisions: patient consent control, FHIR interoperability, audit trail, and even a governed break-glass mode 【31†L838-L847】 . The existence of a break-glass mechanism is notable; ANUJNA should consider a similar feature if not already. WiraChain also enforces RBAC plus audits on chain, which parallels ANUJNA's need for doctor verification. Its approach to off-chain clinical data with on-chain logging suggests ANUJNA's architecture should similarly minimize on-chain PHI.

Other Key Systems, Standards and Platforms

Beyond academic papers, several standards and large-scale systems shape patient-controlled health data sharing:

- **HL7 FHIR & SMART on FHIR:** FHIR (Fast Healthcare Interoperability Resources) is a healthcare data standard (HL7) widely adopted for APIs. It defines *Consent* and *Provenance* resources to encode patient consent and audit trails. SMART on FHIR is an app-launch framework (using OAuth2/OpenID) that lets patient-facing apps connect to EHRs via FHIR APIs. While not a platform, SMART on FHIR underpins many patient apps (e.g. Apple Health Records) by standardizing authentication/authorization scopes. These standards enable ANUJNA's interoperability: e.g. ACTION-EHR and WiraChain both use FHIR data models [6†L250-L259] [30†L347-L355]. FHIR's Consent and OAuth scopes can represent ANUJNA's permissions model.
- **Ayushman Bharat Digital Mission (ABDM) / ABHA (India):** A national digital health initiative launched 2021. Each individual gets an ABHA (Health ID) – a 14-digit or `name@` address – linking all their records. ABDM is built on the *Health Information Exchange and Consent Manager (HIE-CM)* architecture [40†L13-L21] [42†L122-L130]. Health providers (HIPs) and users (HIUs) register through compliant apps. Crucially, **all data exchange is consent-driven**: no health record is shared without the patient's explicit approval [42†L248-L254]. Patients use mobile/web apps to view their records, grant or revoke consent for specific documents, and even provide granular consent (sharing only selected records) [42†L209-L218]. ABDM does *not* store medical data centrally; records remain with the original providers (a federated model) [42†L248-L254]. Its audit model logs consents and accesses via the HIE-CM. There is no formal “break-glass” rule specified (emergency access would still require post-hoc consent/legal basis). *Limitations:* Adoption is still rolling out, and concerns remain about data privacy and ensuring interoperability across diverse systems. **Relevance to ANUJNA:** ABDM is a real-world analog of ANUJNA's vision. Both aim for patient-controlled sharing with a national ID. ANUJNA already includes an ABHA-like ID and the consent workflows. Differences: ABDM is engineering-driven (with government backing) rather than research, and it assumes apps integrate with ABDM APIs. ANUJNA's novelty could lie in specific consent/workflow enhancements (e.g. doctor verification steps, emergency policies) beyond ABDM's public design.
- **Australia's My Health Record (MHR):** A nationwide PHR system (2012 start, opt-out in 2019). Every Australian has a MHR where providers can add summaries, lab reports, prescriptions, etc. Patients can control access via the online portal: e.g. restrict certain providers or mark records as “restricted”. All accesses are logged and viewable by the patient (audit logs). Key aspects: it is *centralized* (data held by the national agency), and the primary consent model is opt-in/opt-out at enrollment, plus patient-set access controls. There is no built-in emergency override; clinicians in urgent care rely on existing consent mechanisms (they can search for a patient's record when needed). *Relevance:* MHR exemplifies a government-mediated PHR (opt-out by default, whereas ANUJNA is opt-in). Its audit and consent controls are comparable to ANUJNA's audit logs. Unlike ANUJNA, MHR does not use distributed ledgers or smart contracts, so ANUJNA's blockchain audit could be more transparent.
- **NHS Digital (UK):** The NHS provides patient online access via the *NHS App* and *GP Connect*. Every UK resident has an NHS number (digital identity). Patients can view some of their GP records through the NHS App. Secondary care (hospitals) is less integrated. A national *Summary Care Record* exists

(accessible across trusts), but patients must opt out if desired. Consent is mostly implied for clinical care and opt-out for data use. There is no national patient-controlled unified record beyond the app. *Relevance:* NHS shows a partial, provider-driven approach. ANUJNA differs in giving **full control** to the patient and using explicit consent protocols for sharing, whereas NHS largely relies on institutional data-sharing agreements. ANUJNA's model is more fine-grained.

- **US CMS/ONC Interoperability (USA):** Recent regulations (21st Century Cures Act, USCDI) require providers to expose patient data via FHIR-based APIs (e.g. the “Patient Access API” and “Provider Directory API”). Patients can download their records and share them with apps. Programs like Blue Button 2.0 allow Medicare beneficiaries to retrieve claims data. Consent here is usually application-based (the patient grants an app access via OAuth). Emergency access is governed by HIPAA (treatment exception); there is no national break-glass system beyond institutional policies. *Relevance:* US standards encourage patient-controlled access (through apps), similar to ANUJNA's goals. However, US systems still often use centralized health IDs (e.g. Medicare#) and don't implement blockchain/auditable trails. ANUJNA's accountable auditing and doctor-verification add layers beyond typical US APIs.

Identified Gaps and Novelty

From this survey, we identify where ANUJNA aligns with or diverges from existing solutions:

- **Existing Overlap:** ANUJNA, like ABDM, Dong 2023, WiraChain, and PCEHRM, centers the patient as data owner with consented sharing. All these systems require patient approval for any data exchange and aim to use modern tech (FHIR, blockchain) to secure it. In particular, ABDM/ABHA already provides the unique health ID and consent framework that ANUJNA proposes. Similarly, ANUJNA's use of FHIR and blockchain resembles ACTION-EHR, WiraChain, and other blockchain-EHR research.
- **Key Differences:** ANUJNA emphasizes **explicit, accountable workflows**: e.g., verifying a doctor's credentials before they join a patient's graph, mandated multi-party signatures for consent, and a central consent manager component. Some surveyed systems (e.g. ABDM, WiraChain) do not require pre-verification of providers beyond system credentials. ANUJNA also intends a formalized “*break-glass*” policy for emergencies (similar to WiraChain), whereas many systems either lack it (e.g. ACTION-EHR, ABDM) or rely on high-level policies. Another difference is ANUJNA's focus on a *unified Health Identity* platform (ANUJNA Health ID) that can span jurisdictions, whereas most projects are national or institutional in scope.
- **Novelty vs. Incremental:** Many surveyed works are engineering prototypes demonstrating feasibility of patient-controlled sharing. ANUJNA's novelty would depend on its **procedural** enhancements: for example, integrating a doctor's verification step into blockchain smart contracts, or its particular audit governance rules. If ANUJNA's core technical stack (FHIR + permissioned blockchain + patient consent) is similar to existing systems, then novelty lies in the *combination and workflow design*, not in fundamental algorithms. For instance, using an “image-based multi-party signature” (as ANUJNA mentions) for key management might be patentable if truly unique. But the literature suggests similar multisig/threshold schemes (e.g., references in ACTION-EHR [8†L810-L817]) are known.

- **Potential Patents and Weaknesses:** Any “image-based signature” idea must be examined against prior cryptographic patents; the surveyed works mostly use standard PKI (public/private keys, symmetric keys). If ANUJNA’s approach is just a novel instantiation of multisig, it may be seen as an engineering variant rather than a new patent. Conversely, if ANUJNA plans unique consent tokens or novel cryptography, those could be patentable.
- **Gaps to Address:** The literature highlights recurring gaps that ANUJNA must address to be publishable/patentable:
 - **Emergency access:** Many systems lack a robust emergency break-glass (ACTION-EHR, ABDM) or treat it as future work. ANUJNA should specify and prototype its emergency workflow (possibly drawing on WiraChain’s model [31†L838-L847]) to claim novelty in that area.
 - **“Right to be forgotten”:** Immutable ledgers conflict with data deletion (GDPR). ANUJNA must address how patient revocation or deletion is handled (this gap is noted in ACTION-EHR [8†L826-L833]). Solutions (e.g. redactable blockchains) might be complex; addressing them could strengthen research value.
 - **Scalability:** Experiments in Dong (2023) showed limits (non-graph DB). ANUJNA should consider how it will scale (perhaps via sharding or graph databases as suggested). Demonstrating performance on large networks would boost publishability.
 - **Interoperability:** While FHIR is used, many systems don’t fully show end-to-end interoperability with real EHRs. ANUJNA’s implementation plan could emphasize conformance to HL7 FHIR US Core/IRI and SMART specifications to stand out as a rigorous engineering contribution.
- **Summary:** ANUJNA largely **duplicates** capabilities of platforms like ABDM or the blockchain prototypes (patient IDs, consent-driven sharing, audit logs) – so its core features per se are **not novel**. The potential **novelties** lie in *process design* (e.g. doctor vetting, consensus policies, UI experience) and any unique crypto protocols. If those are mere engineering additions, the research contribution may be viewed as incremental. To ensure publishability, ANUJNA should highlight any unique policy innovations (e.g. multi-sig consent, cross-jurisdiction interoperability) and rigorously evaluate them. Patenting would require identifying truly novel technical elements beyond known blockchain-EHR solutions.

Novelty and Comparison Matrix

Feature / System	ANUJNA	ABDM / ABHA	SMART on FHIR / EHR Apps	Patient-Centric Blockchain EHR (e.g. ACTION, WiraChain)	Consent Mgmt Platforms (e.g. general OAuth/FHIR consent)	Notes on Novelty
Primary Objective	Unified health ID + consent-driven sharing	Nationwide health ID & record linking	App-based access to provider EHRs	Decentralized patient data sharing using blockchain	Managing patient consent via OAuth, FHIR API	ANUJNA combines ID platform with blockchain sharing.
Identity Model	ANUJNA Health ID (14-digit or email@)	ABHA (14-digit or username@abdm)	Depends on vendor (e.g. login.gov)	Blockchain keys + perhaps OAuth user IDs	OAuth2 tokens / SMART scopes / patient app credentials	Very similar to ABDM; not novel unless new ID scheme used
Consent Model	Explicit patient grant/revoke; multi-party signature; revocation allowed (audit logged)	Explicit patient consent for each record link; granular, revocable [42†L209-L218]	Patient authorizes app via OAuth2 scopes	Patients grant access via blockchain smart contracts [52†L533-L542] ; revocation possible, logged	Standard OAuth consent screens; FHIR Consent resource usage	ANUJNA's multi-party signature and policy enforcement may add innovation.
Access Control	Role-based + user-defined policies; doctor ID verification; blockchain-enforced	Role-based at facilities; patient as owner in ABDM's RBAC extension	Provider/ Patient roles within EHR/ Apps	Smart-contract based (WiraChain extends RBAC with on-chain audits [31†L838-L847])	Attribute/role-based, or DAC via patient's OAuth choice	ANUJNA's stringent verification and blockchain audit layer exceed typical models

Feature / System	ANUJNA	ABDM / ABHA	SMART on FHIR / EHR Apps	Patient-Centric Blockchain EHR (e.g. ACTION, WiraChain)	Consent Mgmt Platforms (e.g. general OAuth/FHIR consent)	Notes on Novelty
Emergency ("Break-Glass")	Controlled mechanism: predefined clinical roles can override with audit trail (to be designed)	Not explicitly defined (emergency HIPAA exceptions apply)	Not standardized (depends on hospital policy)	WiraChain implements break-glass with on-chain logging [31†L838-L847] ; others lacked it [8†L818-L827]	Not part of OAuth/FHIR (external to tech layer)	Designing a robust break-glass procedure could be novel (WiraChain is one example).
Auditability	Immutable ledger of all consents/ transactions, patient and provider actions; patient can view logs	HIE-CM logs consent events; ABDM implies logging in Consent Manager	Some systems log accesses in audit trails (not transparent to patient)	Blockchain provides immutable audit (WiraChain records only minimal event data on chain [31†L838-L847])	Logging via FHIR Provenance or server logs; often opaque to user	ANUJNA's transparent patient-facing audit is stronger than traditional logs.
Data Storage	Off-chain clinical data (encrypted in repository), on-chain metadata/ permissions	Data stays with original providers (federated), linked via pointers	EHR databases per vendor; SMART apps fetch per request	Hybrid: Data off-chain (IPFS or cloud); only hashes/perm on chain [6†L250-L259] [52†L533-L542]	EHR DBs (cloud/on-prem) with federated linking; not blockchain-based	Use of blockchain vs ABDM's federated pointers is a design choice.
Security Model	Permissioned blockchain (Hyperledger) with PKI for identities; multi-sig consent	Centralized gov infrastructure with authentication; uses Aadhaar-like KYC for ABHA	OAuth2 + TLS for app-EHR; assumes EHR security (plus patient auth)	Blockchain immutability + cryptographic signatures (e.g. ACTION-EHR uses PKI) [6†L250-L259]	Relies on centralized security controls in EHR; consent via TLS OAuth	Permissioned blockchain is stronger against insider tampering (similar to other blockchain solutions).

Feature / System	ANUJNA	ABDM / ABHA	SMART on FHIR / EHR Apps	Patient-Centric Blockchain EHR (e.g. ACTION, WiraChain)	Consent Mgmt Platforms (e.g. general OAuth/FHIR consent)	Notes on Novelty
Known Limitations	To be addressed: GDPR deletion, scalability, user experience	Low adoption initially; reliance on smartphone (some digital divide); privacy concerns [42†L248-L254]	Dependency on robust EHR APIs (maturity varies); no cross-system identity	Scalability (non-graph DB issue noted [27†L325-L332]); user onboarding burden	OAuth consent fatigue; often lacks fine-grained controls	ANUJNA inherits common challenges (usability, scalability, etc.). Improvements (e.g. graph DB, user workflow design) may be needed.
ANUJNA vs. Existing	- Already does: Patient ID, consent-driven sharing, audit logs (like ABDM/ WiraChain)
 - Differs: adds doctor credential verification and optional multi-signature consent workflows.
 - Nature of diff: Primarily engineering/ policy enhancements; not a fundamental new technology.	- Already: Unified Health ID with consent-centric exchange.
 - Differs: ANUJNA adds explicit break-glass, multi-party consent logic, and (potentially) new UI/UX flows.
 - Diff: Mostly engineering, building on ABDM's architecture.	- Already: Patients authorize apps to read EHRs.
 - Differs: ANUJNA is cross-provider and blockchain-backed (vs. point-to-point).
 - Diff: Incremental from SMART/ OAuth (more like a new application of the ideas).	- Already: These prototypes use blockchain and patient keys.
 - Differs: ANUJNA's contribution would be in <i>workflow rules</i> (e.g. how consents are collected), wide-scale system integration, and any novel cryptography.
 - Diff: Likely incremental improvements, unless ANUJNA invents a fundamentally new protocol.	- Already: OAuth2/FHIR manage consent to some extent.
 - Differs: ANUJNA's use of distributed ledger for accountability, doctor-verification, and emergency policies goes beyond typical API-based models.
 - Diff: Could be a research/ engineering advance in governance, not core tech.	Overall, ANUJNA largely re-implements known concepts (patient ID + consent + ledger). Its novelty depends on specific workflow, cryptographic policy enhancements. Without a clear novel component, the research may be seen as incremental rather than groundbreaking.

Assessment: Our survey suggests that ANUJNA's core vision — a patient-controlled ID and consent platform — is well-aligned with existing trends (ABDM, blockchain-EHR prototypes). The main research contributions would come from how ANUJNA's mechanisms (doctor verification, multi-sig consent, emergency rules, etc.) are designed, validated, and possibly prototyped. If those are merely engineering choices rather than new

theoretical techniques, their publishability may be limited. We must carefully highlight any *unexplored workflows or cryptographic methods* in ANUJNA to claim novelty. Patentability likewise depends on truly new technical features beyond what's public in the literature.
